Hospital Operations & Billing Intelligence

Page 1: Executive Overview | May 2019 to May 2024

Overview | Billing | Operations | ⟲ Reset | ⓘ Info | ☰ Filters

Showing: All years | All conditions | All insurers | All admission types

Year ▼ | Medical Condition ▼ | Insurance Provider ▼ | Admission Type ▼

Total Admissions	Total Billing	Avg Billing / Admission	Avg Length of Stay	Emergency Admission %	Abnormal Test Result %
50,000	1,277.8M	25,556	15.5 days	32.8%	33.5%
trend	trend	trend	trend	trend	trend

Line chart: Monthly trend

Admissions | Billing | Avg Stay

Axis: Month. Drill down Year > Quarter > Month.
Purpose: show trend and seasonality.
Switch the measure with the buttons above (field parameter).

Donut: Admission Type mix

Elective, Urgent, Emergency share.
Click a slice to cross-filter the page.

Key takeaway (smart narrative)

Auto text: "Emergency admissions are 32.8% of all admissions; average stay is 15.5 days."

Buttons: page navigator, Reset (bookmark), Info (definitions pop-up), Filters (show/hide panel). Hover on any visual opens a tooltip page. 2019 and 2024 are partial years.

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Page 2: Billing & Insurance Analysis

Overview | **Billing** | Operations | ⟲ Reset | ⓘ Info | ☰ Filters

Showing: All years | All conditions | All insurers | All admission types

Year ▼ | Medical Condition ▼ | Insurance Provider ▼ | Admission Type ▼

Total Billing	Avg Billing / Admission	Negative Billing Count
1,277.8M	**25,556**	**96**
trend	trend	flagged, not deleted

Bar: Billing by Insurance Provider

Total | Average

Fields: Insurance Provider, selected measure.

Bar: Avg Billing by Medical Condition

Fields: Medical Condition, Avg Billing. Data bars on values.

Matrix: Insurance Provider x Admission Type

Values: Avg Billing. Conditional formatting colour scale to spot gaps.

Negative billing: refunds and adjustments

Table of Billing Status = Negative. Coral highlight.

Drill through → Details

Interactions: click bar to cross-filter, right-click a condition to drill through to the admission detail page, Total/Average toggle changes the measure.

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Page 3: Patient & Operations Insights

Showing: All years | All conditions | All insurers | All admission types | All age groups

Year ▼ | Medical Condition ▼ | Insurance Provider ▼ | Admission Type ▼ | Age Group ▼

Avg Length of Stay
15.5 days
trend

Emergency Admission %
32.8%
trend

Abnormal Test Result %
33.5%
trend

Column: Stay by Age Group

18-34, 35-49, 50-64, 65+

Bar: Stay by Condition

Avg Length of Stay

What-if: shorter stays

Slider: reduce average stay by [0 —•— 5] days
Sample (1 day): about 50,000 bed-days freed, about 82M billing equivalent (assumes billing is proportional to days). Shows where savings can come from.

Donut: Test Results

Normal, Abnormal, Inconclusive

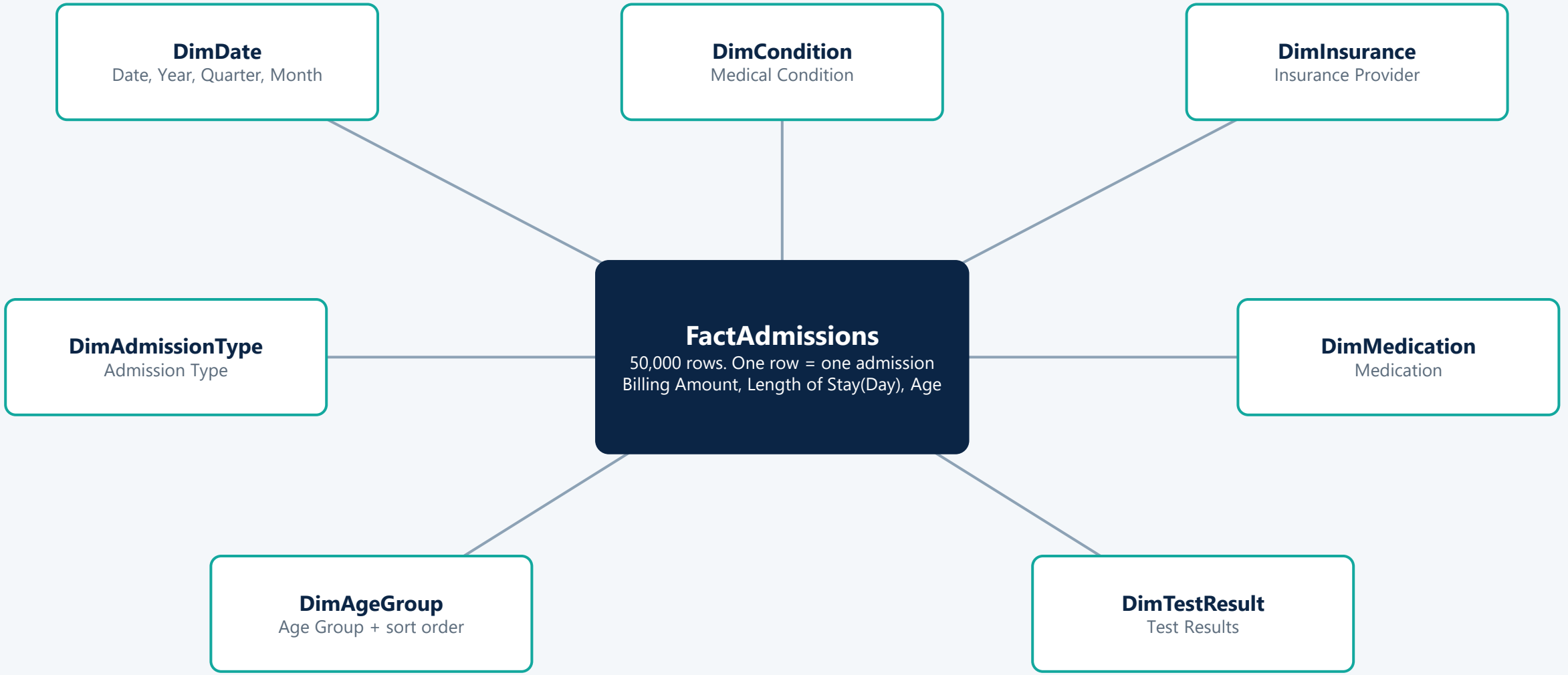
Column: Stay distribution

Length of Stay(Day) 1 to 30

Length of Stay = Discharge Date minus Admission Date (days). What-if parameter adds a scenario measure. Tooltip page shows admissions, billing and average stay.

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Draft data model: star schema



Each Dim (one) to FactAdmissions (many), single direction. DimDate also joins Discharge Date as an inactive relationship. Doctor and Hospital stay hidden in the fact.

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Interactive features and how Power BI builds them

Feature	Where	How in Power BI
Page navigator buttons	All pages	Insert > Buttons > Navigator
Reset filters	Header	Bookmark "Reset" + button action
Info pop-up (definitions)	Header	Bookmark + selection pane overlay
Show / hide filter panel	Header	Bookmarks on slicer group
Metric switch (Admissions / Billing / Avg Stay)	Page 1	Field parameter
Total / Average toggle	Page 2	Field parameter
Drill down Year > Quarter > Month	Page 1	Date hierarchy
Drill through to details	Page 2	Drill-through page
Tooltip page on hover	Pages 1 to 3	Report page tooltip
Dynamic title	All pages	DAX with SELECTEDVALUE
What-if slider	Page 3	What-if parameter
Conditional formatting	Page 2	Colour scale, data bars, coral alerts
Key takeaway text	Page 1	Smart narrative visual
PDF export	Day 8	File > Export > PDF